



FINAL REPORT

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Abstract

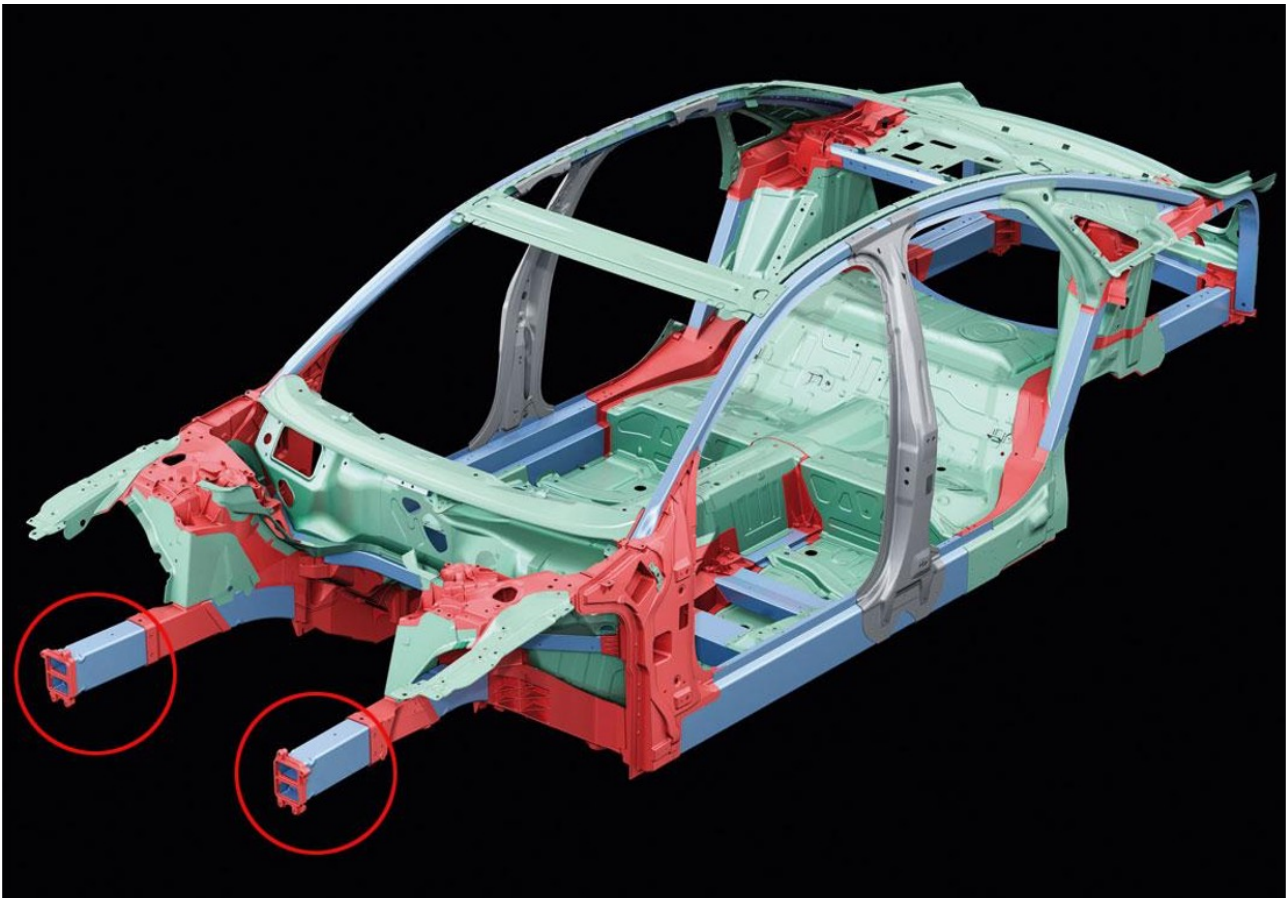
This project aims to determine the Johnson-Cook damage parameters for a tubular structure used in crash-box applications, employing a combination of numerical simulation, experimental data, and optimization techniques. Crash boxes are energy-absorbing components in vehicles, designed to enhance occupant safety by dissipating impact energy through controlled deformation. To achieve this, the Johnson-Cook material model, a widely-used framework in impact simulations, was selected for its ability to predict metal behavior under high strain rates and temperatures. Using ABAQUS for quasi-static simulations and WebPlotDigitizer to extract experimental displacement and force data, the project first establishes a dataset for comparing experimental and numerical results. After conducting simulations for randomly generated sets of Johnson-Cook damage parameters (D1–D5 and a damage evolution parameter), the root mean square error (RMSE) between simulated and experimental force-displacement data is calculated.

An objective function was formulated using a polynomial model, linking the RMSE to the damage parameters. The Jaya Algorithm will be applied to minimize the objective function, identifying the optimal Johnson-Cook damage parameters that best replicate experimental results. This study provides a systematic approach for parameter determination, contributing to improved simulation accuracy for crashworthiness analysis in tubular structures.

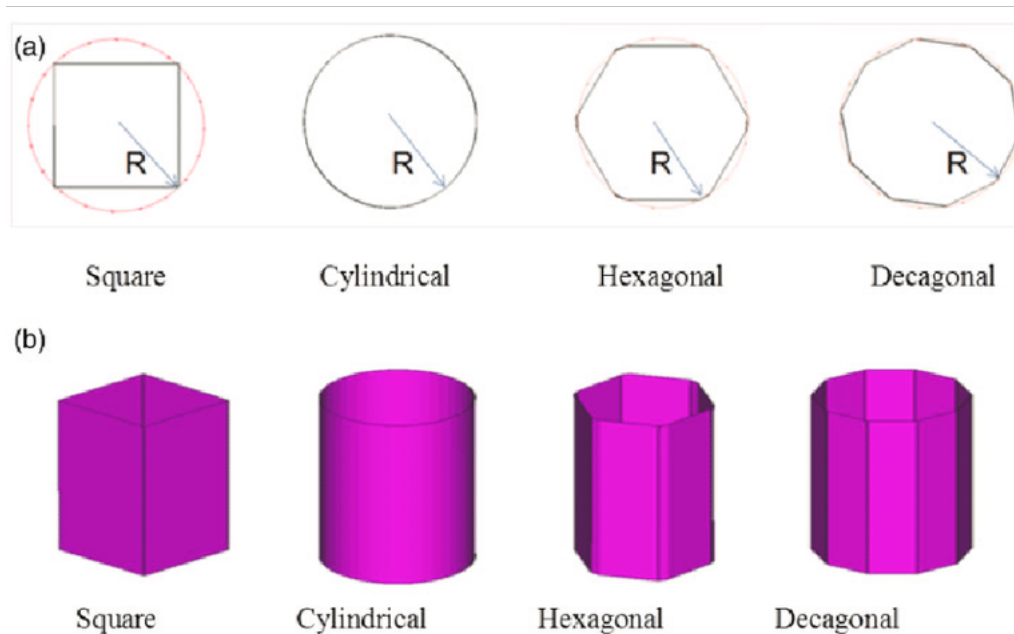
Introduction

A crash box is a specialized energy-absorbing structure, commonly used in automotive and aerospace applications, designed to protect passengers and critical components by dissipating impact energy during a collision. Constructed typically from tubular materials such as aluminum alloys, crash boxes absorb energy through plastic deformation, enabling them to withstand high loads while minimizing the force transmitted to other parts of the structure. The crash-box is used to enhance the safety of the vehicle's occupants by mitigating the effects of collisions. By deforming irreversibly, it dissipates the impact energy, reducing the shock force transmitted to the vehicle's occupants and main structure. This energy absorption is crucial in preventing severe injuries and maintaining the structural integrity of the vehicle during an accident. The crash-box is typically located between the bumper and the main structural frame of the vehicle. It is found at both the front and rear ends of the automobile, serving as the initial point of impact absorption during a collision. This strategic placement ensures that the

crash-box can effectively absorb and dissipate energy before it reaches the main structural components and the vehicle's occupants.



The energy absorption characteristics of crash boxes can be enhanced by incorporating geometric modifications, such as patterned windows or discontinuities, which initiate controlled crushing modes. These structures may collapse in various patterns, including concertina (C) and diamond (D) modes, depending on their design. Research on different section shapes—circular, hexagonal, pentagonal, and more—indicates that such modifications optimize energy absorption efficiency by maximizing energy absorbed (EA), minimizing peak force ($P(\max)$), and improving specific energy absorption (SEA).



Quasi-static and dynamic testing are two key methods for evaluating how well energy-absorbing structures, like crash boxes, perform under different types of loads. Quasi-static testing applies a slow, steady load to the structure at a constant speed, typically a few millimeters per minute, allowing researchers to observe how the material deforms and absorbs energy over time. This slow-loading test helps identify important properties like collapse pattern, force changes, and energy absorption under controlled conditions. Dynamic testing, on the other hand, uses high-speed loads to mimic real-world impact conditions, as in a car crash. This type of test shows how the structure handles sudden impacts, where rapid loading and material response are crucial. Additionally there is a third method called Numerical method in which the researcher uses softwares such as LS-DYNA or ABAQUS to simulate the real world testing methods on a computer and gain information on the tested structure and material. In our project which will be explained in detail towards the end, we use ABAQUS to simulate the quasi-static testing of a tubular structure that has been given in a research paper.

There are many important parameters in deciding which tubular structure is the best for real world applications for a crash-box. We have Energy Absorption, the total energy absorbed during deformation, specific energy absorption which is the energy absorbed per unit mass, peak force, which is the maximum force experienced during the initial phase of deformation, mean crushing force, the average force during the crushing process, structural efficiency, efficiency of the structure in absorbing energy relative to its yield strength and cross-sectional area, crush force efficiency which is the ratio of mean crushing force to the peak crushing force and finally undulations in load-carrying capacity which is the degree of force fluctuation around the mean force.

Triggers, or crush initiators, are small geometric modifications introduced into energy-absorbing structures, like crash boxes, to control how and where they begin to deform during impact. By initiating the crushing process in a controlled manner, these features help create predictable and stable collapse patterns that improve energy absorption. Typical triggers might include notches, grooves, cutouts, or patterned windows strategically placed along the tube's surface. When impact occurs, these initiators act as weak points that direct the deformation to begin at specific locations, reducing the initial peak force ($P(\max)$) and allowing the structure to collapse gradually. This controlled deformation absorbs energy more effectively, which is especially important in crash applications where maintaining a stable, predictable force profile is critical to protecting occupants and important components.

Finite Element Method (FEM) is a powerful computational tool used to analyze the behavior of complex structures under various loading conditions, such as in crash testing of energy-absorbing devices like crash boxes. By breaking down a structure into smaller, manageable elements, FEM allows engineers to simulate how each element responds to stress, strain, and deformation when subjected to forces. This method is especially useful for predicting the performance of materials and structures with complex shapes or designs, where analytical solutions are challenging. Using FEM, engineers can visualize and optimize energy absorption, collapse modes, and stress distribution within the structure before physical testing, saving time and reducing costs. FEM also enables the exploration of different design modifications, such as adding triggers or adjusting geometric shapes, to enhance energy absorption characteristics under both quasi-static and dynamic loading conditions.

I have explained mainly about what a crash-box is in the previous paragraphs but the more important topic is the project that I have been working on using ABAQUS, Excel, Python and WebPlotDigitizer. So in the coming paragraphs I shall give a better insight to what I have been doing these past few months and what the future of the project is as well.

Project Overview

The goal of my project is to determine the Johnson-Cook Damage Parameters of a tubular structure using Jaya Algorithm. In the later paragraphs and steps in this project we will learn about the Johnson Cook

Material model and why it is important and how to find the values of the parameters called D1, D2, D3, D4 and D5 and damage evolution parameter.

Johnson-Cook Material Model:

The Johnson-Cook material model is a widely-used constitutive model that describes how metals respond to different mechanical and thermal conditions, particularly under high strain rates and elevated temperatures. Developed primarily for materials that experience plastic deformation in extreme environments, like during impacts, explosions, or high-speed machining, the model provides a realistic representation of how materials strengthen, soften, and ultimately fail under load. The model combines two components: the Johnson-Cook Strength model, which predicts stress-strain behavior, and the Johnson-Cook Damage model, which estimates the conditions leading to fracture.

The Johnson-Cook Strength model represents the flow stress as a function of strain, strain rate, and temperature. It is expressed as a three-part equation, each part accounting for a key aspect of material behavior. The first term captures strain hardening, where the material becomes stronger as it deforms plastically. The second term accounts for strain rate sensitivity, meaning the material's strength increases with faster deformation rates. Finally, the third term models thermal softening, where the material becomes weaker as temperature rises. By incorporating these factors, the Johnson-Cook Strength model can predict stress-strain behavior over a wide range of conditions, making it suitable for finite element simulations used in industries like automotive and aerospace, where metals often undergo complex loading.

The Johnson-Cook Damage model is used in conjunction with the Strength model to predict material failure. This model tracks damage accumulation as a function of plastic strain, strain rate, and temperature, and it predicts fracture when a critical damage value is reached. The Damage model includes an empirical formula for damage initiation and progression, enabling the simulation of material fracture under dynamic loading, like impacts. Together, the Strength and Damage models provide a comprehensive framework for predicting both deformation and failure. This dual approach is particularly valuable for simulating events like car crashes, where it's crucial to understand not only how materials will deform but also when they will break, providing insights into the overall crashworthiness and safety of structures.

Now we shall get into all the steps of the project that uses this model and other softwares to get towards our goal and achieve our task of finding the best D1-D5 values and the best damage evolution parameter.

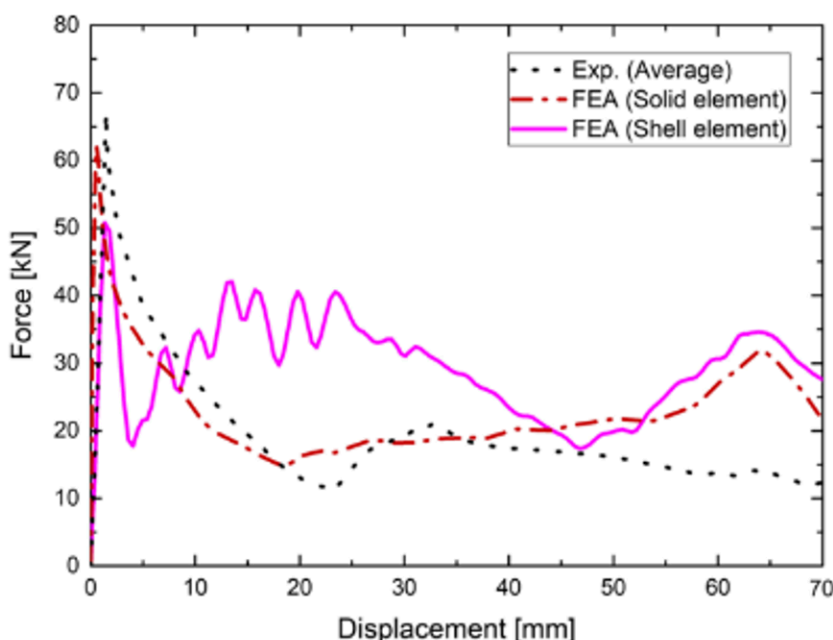
Step 1: Literature Review

The first step in this project was to read a research paper called, “Experimental and numerical investigation on the crashworthiness optimization of thin-walled aluminum tubes considering damage criteria”, by Sinem K. Mert, Murat Demiral, Murat Altin, Erdem Acar and Mehmet A Guler. This paper talked about the performance of thin walled tubes in quasi-static conditions both experimentally and numerically. We looked at both single cell and multi cell tubes in the paper and the CFE and SEA of each were compared to see which is the best thin walled tube in the small sample size of tubes. But, in our case we were only concerned about the single cell tube and how it was numerically tested using Johnson-Cook parameters and other general parameters of any material.

Step 2: Data Extraction (WebPlotDigitizer)

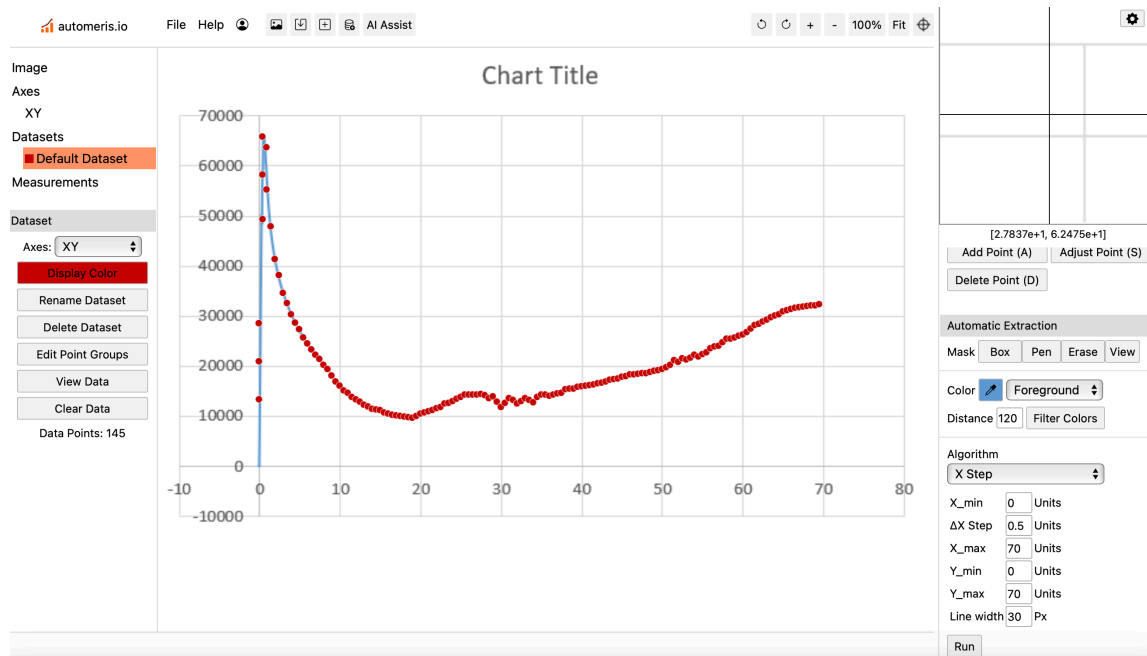
In the next step of the process, the goal is to obtain the experimental average graph for the single cell tube, focusing on its displacement and force data. To do this, WebPlotDigitizer is used to extract data points from the experimental graph. WebPlotDigitizer allows you to manually trace the graph and convert it into a digital dataset by recording the values at each point on the curve.

The displacement in the experiment ranges from 0 to 69.5 mm, and data points are taken at small intervals of 0.5 mm. This detailed interval helps ensure a high level of accuracy when analyzing how force changes with



	A	B
1	exp_disp	exp_force
2	0	0
3	0.5	12.90673
4	1	43.60203
5	1.5	59.25664
6	2	58.18231
7	2.5	52.50367
8	3	48.82024
9	3.5	45.90418
10	4	43.1416
11	4.5	40.99293
12	5	39.61164

displacement. For each interval point, the corresponding force was recorded, resulting in a complete dataset that tracks the force response as displacement increases. These displacement and force values were then saved in an Excel file, which allows for further analysis and processing. This dataset will be essential for comparing experimental results to simulated results.

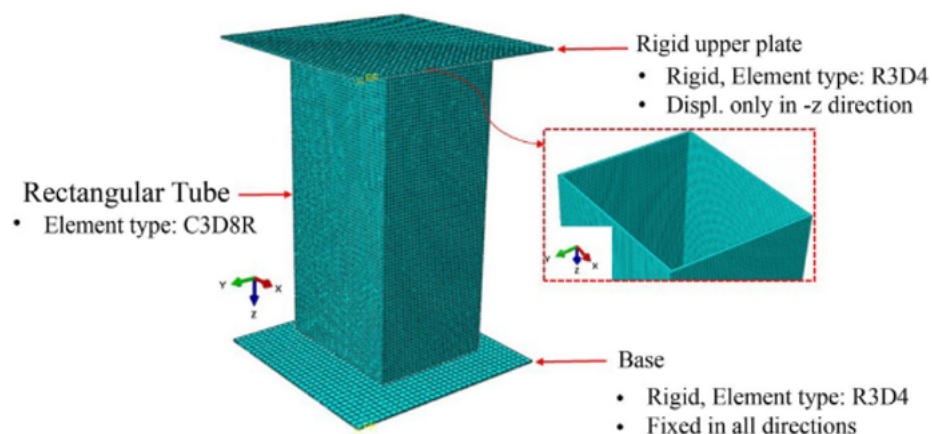


Step 3: Johnson-Cook Model Setup

The Johnson-Cook material model is widely used to predict how metals behave under high strain rates and temperatures, making it especially useful in simulations of impacts, crashes, and machining processes. It consists of two main parts: the Johnson-Cook Strength model, which describes how materials respond to stress, strain, strain rate, and temperature, and the Johnson-Cook Damage model, which predicts when and where materials will fracture. The Strength model provides a formula for how materials harden with deformation and respond to higher strain rates or temperatures, while the Damage model calculates damage accumulation leading to failure. Together, these models help engineers simulate material deformation and failure under extreme conditions, enabling accurate design and analysis of structures exposed to high-impact forces.

We are more concerned about the damage model as we are looking to find the values D1-D5 for a tubular structure by using the data and information we gather throughout this project using softwares that enable us to do this task. The Johnson-Cook Strength model values of the structure is given to us in the paper that was mentioned above, we will be copying those values for our later simulation work but for the values of D1-D5 and the damage evolution parameter, we will first create 150 different randomly generated sets of D1-D5 and damage evolution parameter values in an excel file. D1-D5 have a constraint to be between -10 to 10, while the damage evolution parameter has a constraint to be between 0 to 1.

Then we use ABAQUS to create the single cell tubular structure given in the paper mentioned above in step 1. We use all the values that have been given in the paper except the values for D1 to D5 and the damage evolution parameter. With this now we conduct the simulation for each of the 150 sets of values that have been randomly generated using the same tubular structure and we will look into the information gained more in the later steps.



Step 4: Simulation and Error Calculation

After the simulation of each and every set of data that was randomly generated, we will have to find the force vs displacement graph and values. The values can be easily found in ABAQUS but to get it in the format of the experimental average in which the displacement values were in the intervals of 0.5mm from 0 to 69.5mm and with their corresponding force values we have to do some things. First we shall take the force vs displacement table and in a new excel file create a line graph using scatter plot for the data given. Then take a screenshot of the graph and save it in any folder. Use this picture in WebPlotDigitizer and create a table of the similar format of experimental average.

Then we will have to use Python, I used the Scikit-learn library to write my program to find the error between the experimental average and the simulated tubular structure. I wrote a program to find the root mean square error between the two datasets available at a time. This needs to be done for all 150 different sets of Johnson-Cook damage parameters. All these error values will be stored with their respective sets in the excel file where the 150 sets of values were generated.

	A	B
1	0	0
2	0.5	62.62877
3	1	59.89428
4	1.5	46.69739
5	2	40.39617
6	2.5	37.30501
7	3	33.97606
8	3.5	32.07381
9	4	30.40934
10	4.5	28.2693
11	5	27.08039

	A	B
1	exp_disp	exp_force
2	0	0
3	0.5	12.90673
4	1	43.60203
5	1.5	59.25664
6	2	58.18231
7	2.5	52.50367
8	3	48.82024
9	3.5	45.90418
10	4	43.1416
11	4.5	40.99293
12	5	39.61164

exp_disp	exp_force	fea_disp	fea_force
0	0	0	0
0.5	12.90673	0.5	62.62877
1	43.60203	1	59.89428
1.5	59.25664	1.5	46.69739
2	58.18231	2	40.39617
2.5	52.50367	2.5	37.30501
3	48.82024	3	33.97606
3.5	45.90418	3.5	32.07381
4	43.1416	4	30.40934
4.5	40.99293	4.5	28.2693
5	39.61164	5	27.08039

```

1  import pandas as pd
2  from sklearn.metrics import mean_squared_error
3  import math
4
5  file_path = 'Error_1.csv'
6  data = pd.read_csv(file_path)
7
8  print(data.head())
9  print(data.columns)
10
11 fea_force = data["fea_force"]
12 experimental_force = data["exp_force"]
13 mse = mean_squared_error(experimental_force, fea_force)
14 rmse = math.sqrt(mse)
15
16 print(f"Root Mean Squared Error between FEA and Experimental force values: {rmse}")

```

```

• (venv) PS C:\Sreeram\SOP\jaya1> python error.py
  exp_disp  exp_force  fea_disp  fea_force
0      0.0    0.000000      0.0    0.000000
1      0.5   12.906726      0.5   62.628766
2      1.0   43.602035      1.0   59.894275
3      1.5   59.256642      1.5   46.697388
4      2.0   58.182306      2.0   40.396171
Index(['exp_disp', 'exp_force', 'fea_disp', 'fea_force'], dtype='object')
Root Mean Squared Error between FEA and Experimental force values: 10.390476669219161

```

Current Work

The previous steps to the project are either done or ongoing, while the steps of the project I will be talking about in this stage is something that still has to be done.

Step 5: Formation of Objective Function

I have read a few papers on how to use the data I have been getting from the simulations I have been doing on the ABAQUS software and the most popular one I could say is to equate the error value to a polynomial which includes all the constants D1 to D5 an damage evolution parameter with a certain number of coefficients.

To explain the whole process more clearly, first, we need to have the RMSE values of 28 different simulations for 28 different randomly generated sets of constants for the damage parameters. These values will be used in the later stages of this step.

Next we will be making the polynomial for our objective function, so what we will be doing is, using the equation $OBJ(x_i) = a_0 + a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_6x_6 + a_7x_1^2 + \dots + a_{12}x_6^2 + a_{13}x_1x_2 + \dots + a_{27}x_5x_6$. Here $OBJ(x)$ is the error value for that set of constants, $x_1 - x_6$ are the constants D1 - D5 and damage evolution parameter respectively. We will be making 28 different equations for the 28 different sets with their corresponding error values.

Now with all these equations, we will be solving them using a python program that I have written. Hence this will get us the values of the 28 different coefficients a_0 to a_{27} . Thus, our objective function will be formed.

$$OBJ(x) \cong a_0 + \sum_{i=1}^n a_i v_i + \sum_{i,j=1}^n b_{ij} v_i v_j$$

Step 6: Jaya Algorithm Optimization

The final step is to create a program using python or any other language that mimics Jaya algorithm which will be our method to optimizing the objective function that was formed in the last step. We will use this algorithm to minimize this function and obtain the damage parameters that we aimed to find. This shall be the best values of D1 - D5 and damage evolution parameter that can be used numerically for this specific experiment as it will show the least error with the experimental graph that was taken in the previous steps.

Conclusion

In this project, we explored the intricate process of determining the Johnson-Cook Damage Parameters (D1-D5) for a tubular structure used in crashworthiness applications. By employing a combination of experimental data, numerical simulations, and optimization techniques, we aimed to find the most accurate damage parameters that could simulate the material behavior of thin-walled tubes under quasi-static conditions.

The project began with an in-depth analysis of existing research and the extraction of experimental data, followed by the use of Abaqus to simulate the behavior of the tubular structure under various conditions. By generating multiple sets of Johnson-Cook damage parameters, we were able to conduct simulations and compare the resulting force-displacement graphs with experimental data using WebPlotDigitizer. The root mean square error (RMSE) between the experimental and simulated results was calculated for each set of parameters, forming the basis for optimization.

Looking ahead, we will use a method of polynomials to form an objective function that will be optimized using the Jaya Algorithm. This optimization process aims to minimize the error between the experimental and simulated results, ultimately providing the most accurate set of damage parameters for the specific tubular structure being studied.